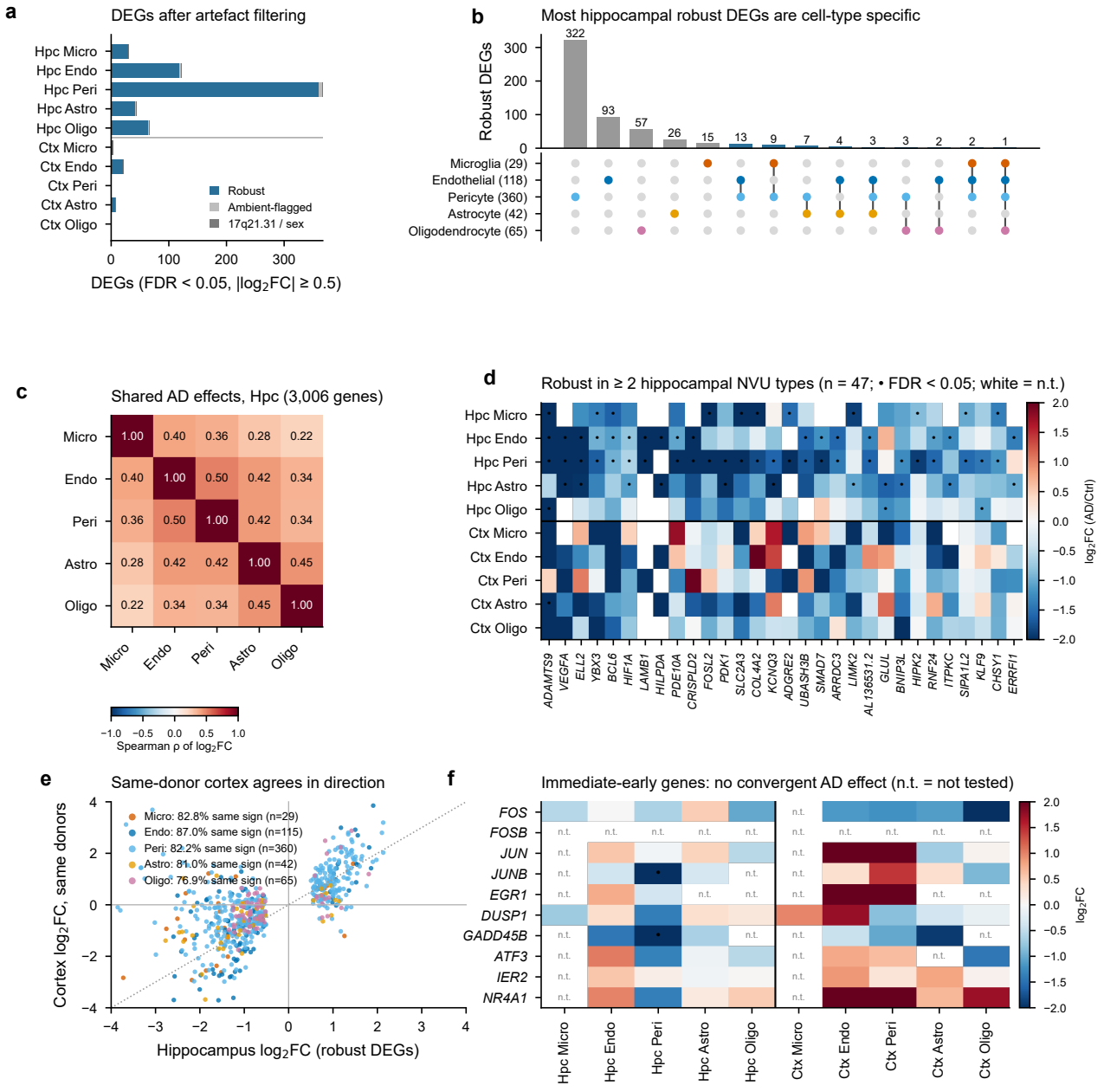

a
DEGs after artefact filtering
Hpc Micro
Hpc Endo
Hpc Peri
Hpc Astro
Hpc Oligo
Ctx Micro
Ctx Endo
Ctx Peri
Ctx Astro
Ctx Oligo
Robust
Ambient-flagged
17q21.31 / sex
0
100
200
300
DEGs (FDR < 0.05, |log2FC| ≥ 0.5)
b
Most hippocampal robust DEGs are cell-type specific
Robust DEGs
322
93
57
26
15
13
9
7
4
3
3
2
2
1
Microglia (29)
Endothelial (118)
Pericyte (360)
Astrocyte (42)
Oligodendrocyte (65)
c
Shared AD effects, Hpc (3,006 genes)
Micro
Endo
Peri
Astro
Oligo
1.00 0.40 0.36 0.28 0.22
0.40 1.00 0.50 0.42 0.34
0.36 0.50 1.00 0.42 0.34
0.28 0.42 0.42 1.00 0.45
0.22 0.34 0.34 0.45 1.00
−1.0 −0.5 0.0 0.5 1.0
Spearman ρ of log2FC
d
Robust in ≥ 2 hippocampal NVU types (n = 47; • FDR < 0.05; white = n.t.)
Hpc Micro
Hpc Endo
Hpc Peri
Hpc Astro
Hpc Oligo
Ctx Micro
Ctx Endo
Ctx Peri
Ctx Astro
Ctx Oligo
ADAMTS9
VEGFA
ELL2
YBX3
BCL6
HIF1A
LAMB1
HILPDA
PDE10A
CRISPLD2
FOSL2
PDK1
SLC2A3
COL4A2
KCNQ3
ADGRE2
UBASH3B
SMAD7
ARRDC3
LIMK2
AL136531.2
GLUL
BNIP3L
HIPK2
RNF24
ITPKC
SIPA1L2
KLF9
CHSY1
ERRFI1
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
log2FC (AD/Ctrl)
e
Same-donor cortex agrees in direction
Micro: 82.8% same sign (n=29)
Endo: 87.0% same sign (n=115)
Peri: 82.2% same sign (n=360)
Astro: 81.0% same sign (n=42)
Oligo: 76.9% same sign (n=65)
Cortex log2FC, same donors
Hippocampus log2FC (robust DEGs)
f
Immediate-early genes: no convergent AD effect (n.t. = not tested)
FOS
FOSB
JUN
JUNB
EGR1
DUSP1
GADD45B
ATF3
IER2
NR4A1
n.t.
Hpc Micro
Hpc Endo
Hpc Peri
Hpc Astro
Hpc Oligo
Ctx Micro
Ctx Endo
Ctx Peri
Ctx Astro
Ctx Oligo
log2FC